

TaxaWizard

Conversational workflow designer for the TaxaID ecosystem. An LLM-powered interview identifies your inputs, parameters, and pipeline steps, then generates a self-contained R script, methods text, or Shiny application.

The TaxaID ecosystem has many possible workflows, and choosing the right combination of packages and parameters can be daunting. TaxaWizard makes this easier through a guided conversation: describe your data and goals, and it generates the appropriate script. `workflow_app()` can also convert any R script (not just TaxaID scripts) into a Shiny application with point-and-click inputs.

Installation

```
install.packages(c("httr2", "jsonlite")) # required
install.packages("shiny")               # optional, for browser/viewer mode
devtools::install("path/to/TaxaWizard")
```

Requires an LLM API key (Anthropic by default). See the TaxaTools API Setup vignette for configuration.

Quick Start

```
library(TaxaWizard)

# Launch the interview (auto-selects console, browser, or RStudio viewer)
workflow_create()

# Or specify a mode
workflow_create(mode = "browser") # opens in web browser
workflow_create(mode = "console") # readline in terminal
workflow_create(mode = "viewer")  # RStudio Viewer pane
```

The assistant asks about your data type, available inputs, and analysis goals, then generates a script tailored to your path through the TaxaID pipeline. Supported workflow paths include:

- **eDNA / metabarcoding** — DADA2 seqtab or FASTA → BLAST → likelihood model → Bayesian or LLM assignment
- **Acoustic** — BirdNET-Analyzer CSV output → match data; or Xeno-canto reference recordings + BirdNET → acoustic likelihood model
- **Camera trap / image** — Animl, iNaturalist CV, or Wildlife Insights output → match data; or labeled reference images → image likelihood model
- **Reference library building** — taxa names (from TaxaExpect or user-supplied) → site-specific NCBI reference library; or load a local CRABS or FASTA database
- **Occurrence-based priors** — taxa + location → GBIF occurrences → habitat → spatially explicit priors
- **Assignment convergence** — any combination of the above → likelihoods + priors → posteriors → consensus → report

When the interview is complete, TaxaWizard generates:

- **R script** with checkpoint/resume, error recovery, and debug mode
- **Methods text** summarizing the workflow for manuscripts
- **Shiny app** (via `workflow_app()`) for point-and-click execution

Fixing Errors in Generated Scripts

```
# If a generated script errors, resume the conversation with context
workflow_fix()
```

Converting Scripts to Shiny Apps

```
# Convert any TaxaWizard-generated (or annotated) script to a Shiny app
workflow_app("my_workflow.R")

# Annotate a generic R script first, then convert
annotate_script("my_analysis.R")
workflow_app("my_analysis.R")
```

Key Functions

Function	Purpose
workflow_create()	Launch interactive interview (main entry point)
workflow_fix()	Resume after script error with diagnostic context
workflow_app()	Convert generated script to Shiny app
annotate_script()	Annotate generic R scripts for Shiny conversion
workflow_engine()	Stateless LLM engine (advanced / programmatic use)

Part of TaxaID

TaxaWizard sits outside the TaxaID dependency chain. It reads package metadata files and generates scripts that call the other TaxaID packages – it does not import them directly.

See the TaxaID README for ecosystem overview and installation instructions.

Citation

Lafferty, K.D., 2026, TaxaID – A modular R ecosystem for Bayesian taxonomic assignment: U.S. Geological Survey software release, <https://doi.org/10.5066/xxxxxx>.

Software Requirements

- R ($\geq 4.1.0$)
- An LLM API key (Anthropic recommended) is required for the conversational engine
- shiny (for browser/viewer chat interface and generated apps; in Suggests)

All dependencies are declared in the DESCRIPTION file and installed automatically.

Developed with Claude Code (Anthropic).